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A System and Method for Voice-Driven Semantic Image Search Leveraging AWS and OpenAI

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Table of Contents

1 Abstract

2 Introduction

3 Literature Survey

4 Problem Definition

5 Proposed Work

6 Methodology & Implementation

7 Results & Discussions

8 Conclusion

9 References

ABSTRACT

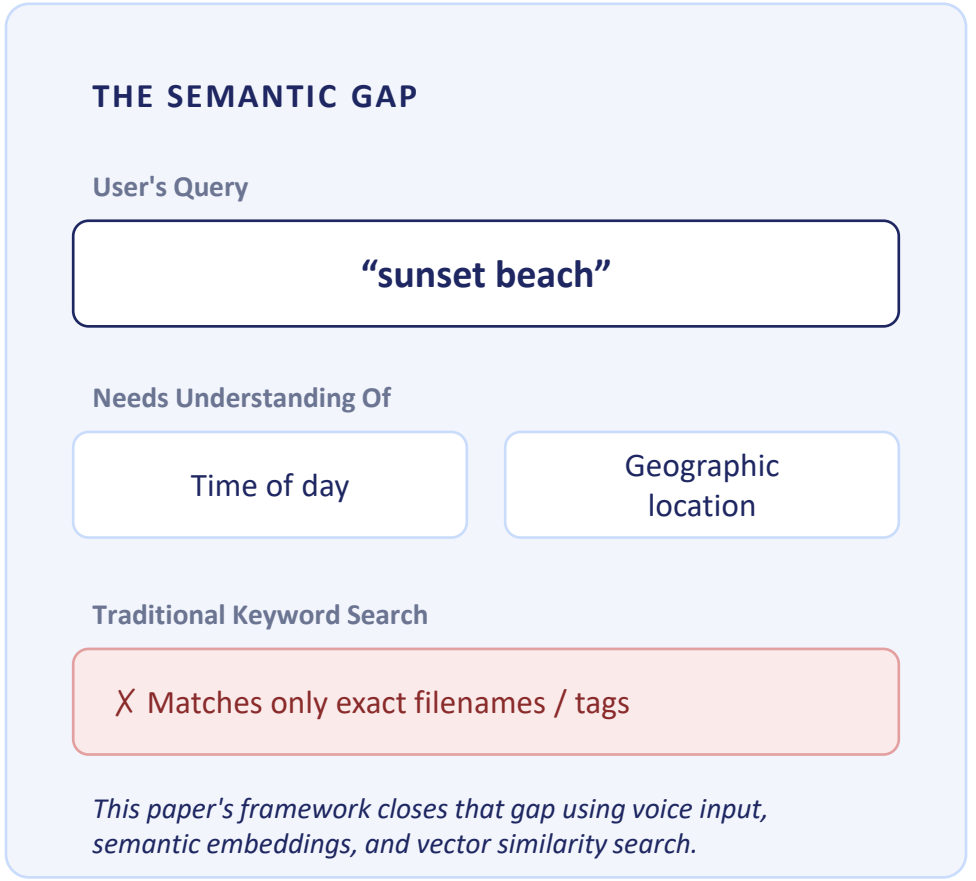
The Paper at a Glance

The massive growth in personal digital photography has made it hard to retrieve photos efficiently — traditional gallery apps depend on metadata and miss the semantic relationships between images. This paper introduces a cloud-based application that searches photos using natural-language voice input and semantic meaning.

- **Approach:** Combines AWS Rekognition (computer vision), OpenAI text-embedding-3-small (semantic embeddings), and Qdrant (vector similarity search).
- **Dataset:** Evaluated on 1,247 personal photographs across 15 semantic categories.
- **Headline Results:** 795 ms mean search latency and 86% Precision@10 — well above the keyword-search baseline.
- **Key Feature:** Confidence-based ranking and cross-modal understanding — a query like “red sports car” retrieves semantically relevant images, not just exact keyword matches.
- **Contribution:** A practical reference architecture for incorporating cloud AI services into mobile applications for personal photo management.

Why Personal Photo Search Is Broken

- The average smartphone user takes about **1,500 photos every year**, creating huge personal image archives that are hard to organize or search.
- Traditional gallery apps rely on **metadata — filenames and manual tags** — not on what is actually inside the picture.
- This creates a **“semantic gap”** between low-level visual data and the way people actually describe what they remember.
- Example: searching “red cars” should return **any red car**, regardless of make or filename — traditional search can’t do this.



What Existing Systems Cannot Do

Traditional vs. Proposed Approach

TRADITIONAL SYSTEMS

- ✗ Search relies on exact keyword / filename matching
- ✗ No understanding of meaning or context in a query
- ✗ Manual tagging required — tedious, not scalable [4]
- ✗ Cannot interpret compound ideas (e.g., time + location)
- ✗ No transparency into why a result was returned

PROPOSED FRAMEWORK

- ✓ Natural-language voice queries, spoken like everyday speech
- ✓ OpenAI embeddings capture semantic meaning of query & image
- ✓ AWS Rekognition auto-labels images — no manual tagging
- ✓ Vector similarity search (Qdrant) retrieves conceptually related photos
- ✓ Confidence score (0–100%) shown with every result

Research gap: no existing system unifies voice input, semantic understanding, and image analysis with a transparent, confidence-scored result set.

Core Contributions of This Work

1 Voice-First Interface

A natural-language, voice-driven interface for searching personal photo libraries — no typing required.

2 AWS Rekognition Integration

Automated, complete extraction of visual features: objects, scenes, text, and celebrities.

3 OpenAI Semantic Search

text-embedding-3-small converts query and image descriptions into 1536-dim vectors that capture meaning, not just words.

4 Confidence-Based Ranking

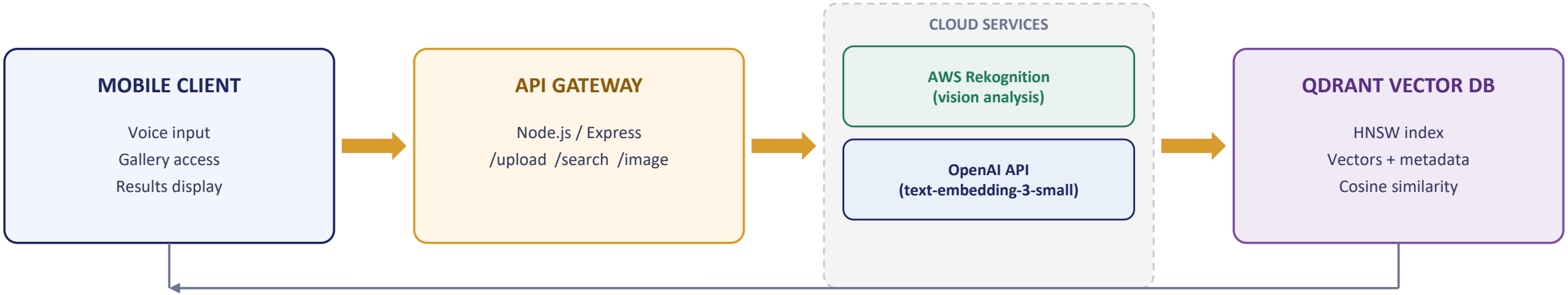
Every result is returned with a transparent relevance score, so users know how much to trust it.

Prior Approaches to Image Retrieval

Approach	Key Idea	Limitation / Gap
Content-Based Retrieval (QBIC, 1990s) [3]	Color histograms, texture & shape descriptors for similarity search	Semantic gap — visual similarity ≠ conceptual meaning
Deep Learning Vision (AlexNet, ResNet, Faster R-CNN) [5][6]	Automatic object / scene feature extraction, human-level accuracy	Needs a production-ready service — provided here by AWS Rekognition
Text Embeddings (Word2Vec, BERT, OpenAI) [7][8][9]	Represent meaning as vectors; text-embedding-3-small yields 1536-dim vectors	Embeddings alone don't search personal images end-to-end
Vector Databases (Qdrant, Pinecone, Weaviate, Milvus) [10][11]	HNSW indexing gives log-time similarity search at scale	Trade-offs between latency, recall and index size across systems
Multimodal Retrieval (CLIP) [13]	Shared image-text embedding space, trained on 400M pairs	No system combines voice + semantic search + confidence scoring — the gap this paper fills

Full reference list is provided at the end of the paper.

Block Diagram of the Framework



Four-layer, cloud-native microservices architecture: presentation → API gateway → AI processing → vector storage.

Presentation layer: Flutter cross-platform mobile client with built-in speech recognition.

API layer: Node.js / Express routes and coordinates all requests.

Processing layer: AWS Rekognition (vision) + OpenAI (embeddings) analyze content.

Storage layer: Qdrant vector database performs the similarity search.

Two Core Pipelines

A. Image Processing & Indexing Pipeline

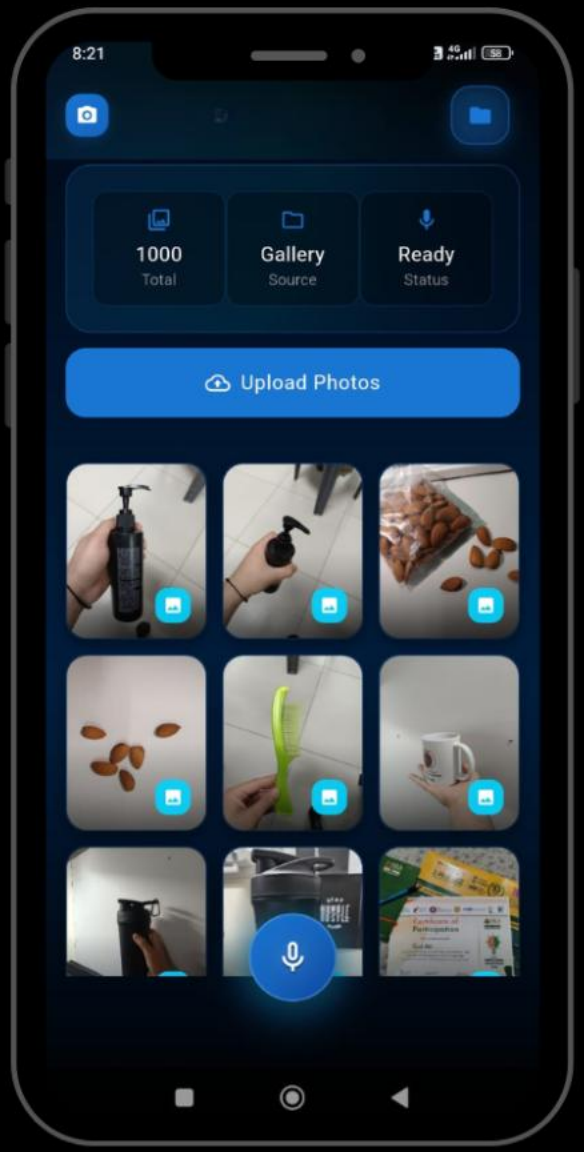
1. Validate format & size
2. AWS Rekognition: objects, text, faces
3. Convert labels to structured text
4. OpenAI: generate 1536-dim embedding
5. Store in Qdrant, index with HNSW

B. Voice Search & Retrieval Pipeline

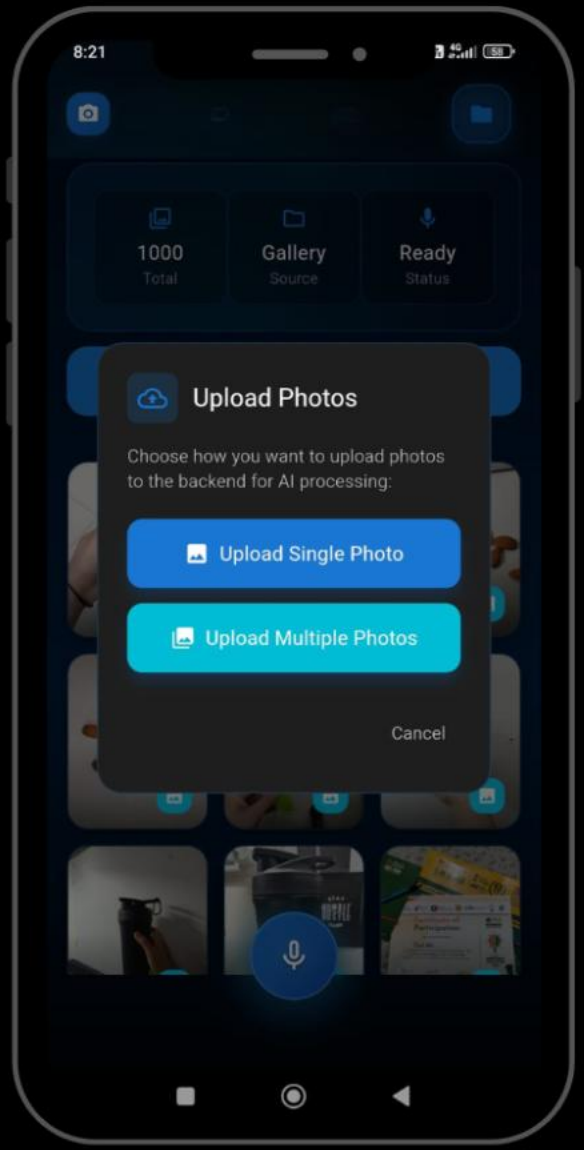
1. Speech-to-text on device
2. OpenAI converts query to embedding
3. Qdrant cosine search → top 20 matches
4. Convert scores to confidence % (0-100)

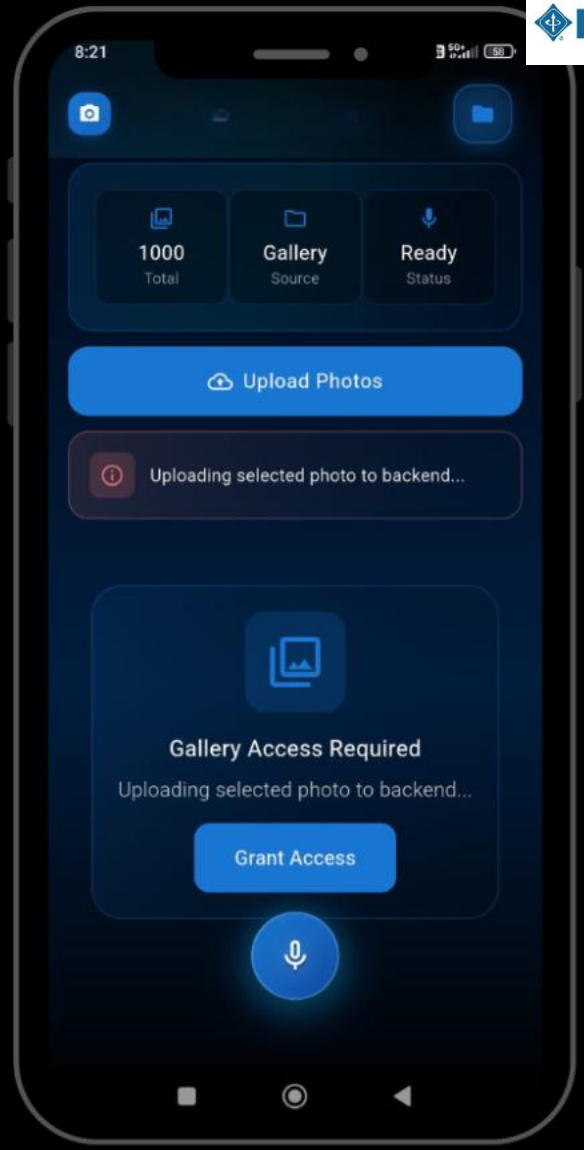
Confidence bands applied to every result:

- **Excellent** ≥ 90%
- **Good** 70 - 89%
- **Fair** 50 - 69%



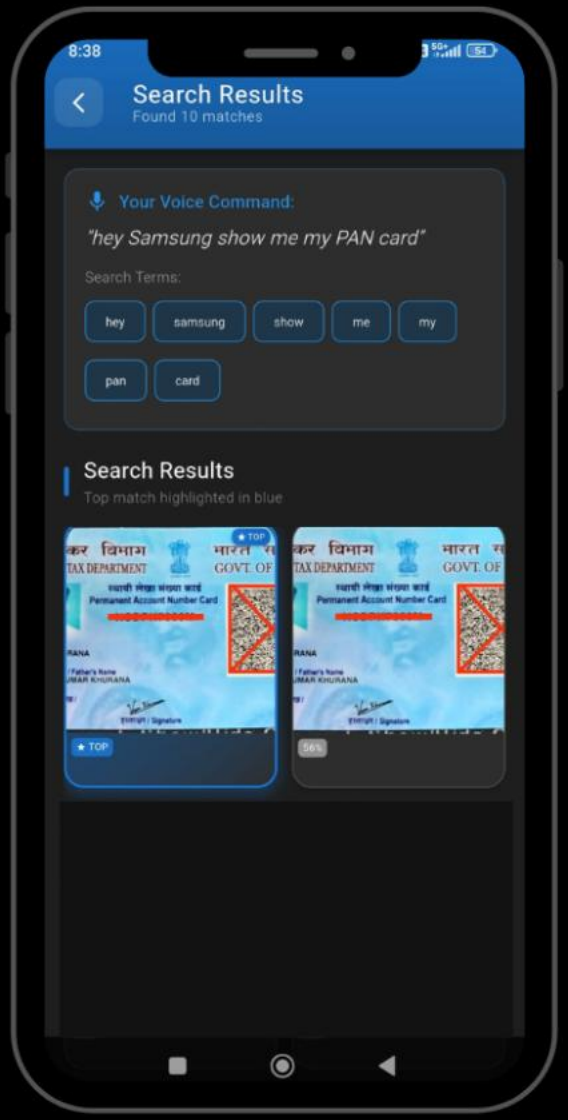
App's Dashboard



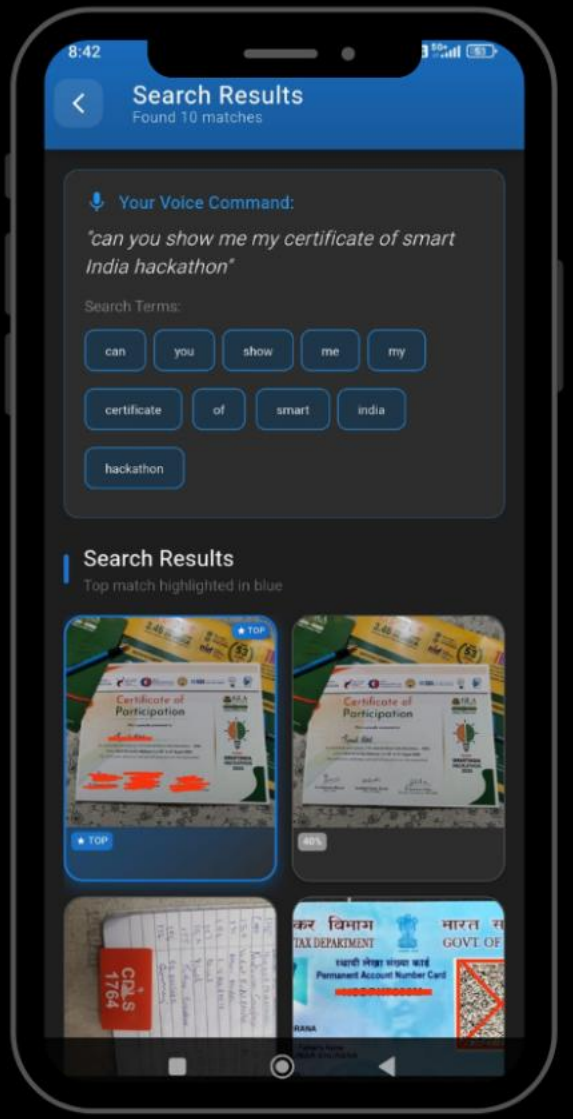


Upload Feature

We have assumed that our app has full access to the user's gallery.



The user asked for his/her PAN Card and these are the top results



The user asked for his/her Smart Indian Hackathon Certificate and these are the top results.

The search results can be managed by adjusting the threshold value inorder to remove the irrelavant matches.


[redacted]
1000
Total
Gallery
Source
Listening
Status
Upload Photos
IEEE

Experimental Setup

Dataset	Test Queries	Test Device	Backend Host
1,247 personal photographs across 15 semantic categories (cars, people, food, sunsets, animals, beaches, text, etc.)	50 natural-language queries: object, scene, activity, attribute & text types	Samsung Galaxy S21 8 GB RAM	Render platform 512 MB RAM

Frontend Stack	Backend Stack	AWS Rekognition	Vector Database
Flutter 3.16 speech-to-text + image-picker	Node.js 18, Express, aws-sdk 3, openai, qdrant-client	Region: us-east-1	Qdrant Cloud (managed hosting)

Semantic Retrieval Performance

86%
Precision@10
(vs 52% keyword baseline)

0.83
Recall@10

0.84
F1-Score

795 ms
Mean search latency

Precision@10 by Query Type

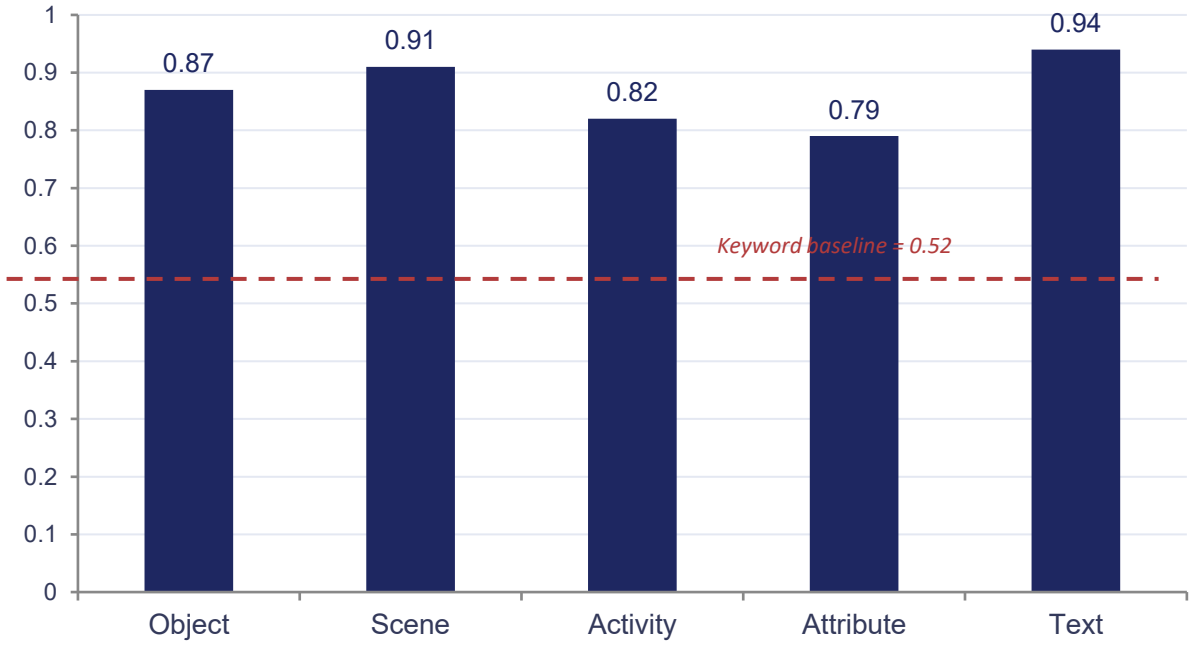


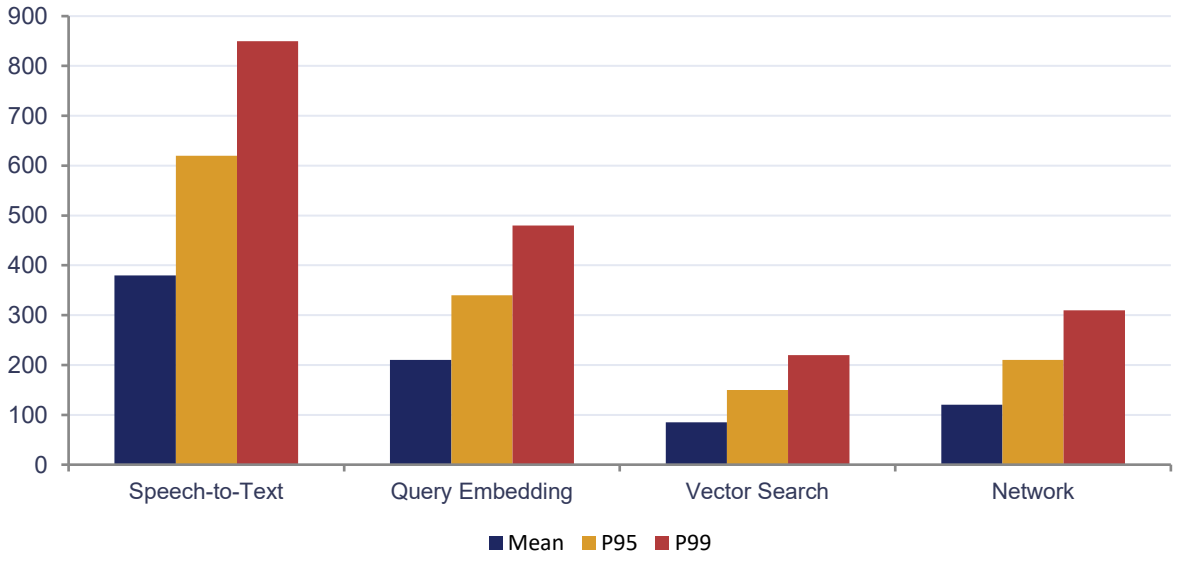
Table III — Semantic Retrieval Performance by Query Type

Query Type	Queries	Prec@10	Recall@10	F1
Object	15	0.87	0.83	0.85
Scene	12	0.91	0.88	0.89
Activity	10	0.82	0.78	0.80
Attribute	8	0.79	0.74	0.76
Text	5	0.94	0.92	0.93

Confidence reflects trust: relevant results averaged 0.81 confidence, vs 0.43 for irrelevant ones — a strong signal for the user.

Performance vs. Existing Systems

End-to-End Latency Breakdown (ms)



Total mean latency: 795 ms • P95: 1,320 ms • P99: 1,860 ms (Speech-to-text dominates at 47.8% of total)

Table V — vs. Existing Gallery Applications

Feature	Google Photos	Apple Photos	Proposed
Keyword Search	Yes	Yes	**Yes**
Voice Search	Limited	No	**Yes**
Semantic Understanding	Partial	No	**Yes**
Confidence Scoring	No	No	**Yes**
Real-time Indexing	Yes	Yes	**Yes**

84% F1-score with off-the-shelf commercial APIs vs. 87% for a custom CLIP-based system — a 3-point trade-off for far faster development and simpler infrastructure.

Voice recognition: 8.1% overall word error rate, 92.9% command success. Error rate rises with query length — 4.2% (2-3 words) → 7.8% (4-6 words) → 12.4% (7+ words).

Cloud vs. on-device: cloud-based embedding generation was 4.2× faster and 23% more accurate than on-device processing.

CONCLUSION

Limitations & Future Work

LIMITATIONS

- Dataset of 1,247 images is smaller than typical user galleries (5,000–20,000)
- Voice recognition supports English only
- Cold-start indexing takes 47 minutes for 1,247 images
- Sending personal photos to cloud APIs raises privacy concerns

FUTURE WORK

- Large-scale, longitudinal evaluation on real user galleries
- Multilingual and accented voice recognition support
- On-device / privacy-preserving embedding & processing
- Progressive & incremental indexing to reduce cold-start time
- Multimodal (CLIP) retrieval and personalized ranking via user feedback

CONCLUSION

A Practical Blueprint for Semantic Photo Search

This paper presented a voice-driven semantic image search framework combining AWS Rekognition, OpenAI text embeddings, and Qdrant vector similarity search — enabling natural, conversational retrieval of personal photographs.

86%	795 ms	0.84	1,247
Precision (vs 52% baseline)	End-to-end search latency	F1-Score	Photos in test dataset

Key takeaway: off-the-shelf cloud AI services — without custom multimodal training — can deliver transparent, confidence-ranked semantic search at production-ready latency.

Thank You

Questions & Discussion

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